

LASSO-Based Classification of Earthquake Events with Fiber Optic Distributed Acoustic Sensors

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Abstract—In this manuscript, a dataset collected through Distributed Acoustic Sensors for earthquake event classification is leveraged. The application of the Least Absolute Shrinkage and Selection Operator model has resulted in an accuracy rate of 99.58%. This performance underscores the advantages of utilizing DAS data, showcasing its potential as a robust and effective tool for earthquake event classification. These findings highlight the promising prospects of DAS technology in seismic monitoring and event classification, offering a valuable contribution to the field of earthquake detection and characterization.

Index Terms—Distributed Acoustic Sensing, Seismic Event Classification, LASSO, Principal Component Analysis, Deep Learning

I. INTRODUCTION

Distributed Acoustic Sensors (DAS) harness the Rayleigh backscattering phenomenon to detect and analyze acoustic changes with good precision. In Rayleigh backscattering, a small fraction of incident light in a fiber optic cable is scattered backward when it encounters microscopic variations in the fiber's refractive index. Interaction of acoustic waves with the optical fiber induces minute changes in the refractive index, causing detectable fluctuations in the Rayleigh backscattered light pattern [1].

DAS systems continuously monitor these variations along the entire length of the fiber, effectively turning it into a distributed sensor. These variations not only affect the intensity of the backscattered light but also introduce a phase difference. The phase difference $\Delta\psi$ caused due to the acoustic strain P , and thereby the refractive index change, Δn is given as [2],

$$\Delta\psi = \beta L \left(\frac{\Delta L}{L} \right) + L \left(\frac{\partial \beta}{\partial n} \right) \Delta n + L \left(\frac{\partial \beta}{\partial \alpha} \right) \Delta \alpha \quad (1)$$

Here, L indicates the gauge length of the fiber, β is the propagation constant of the light signal, α is the diameter of the fiber core. Accordingly, if E is the Young's modulus of the core of the fiber, the fiber's phase sensitivity is represented as [2],

$$\text{Phase Sensitivity} = \frac{\Delta\psi}{P} \frac{E}{\beta L} \quad (2)$$

This technology allows for high-resolution, real-time detection of acoustic events by capturing both intensity and phase differences in the scattered light. It is invaluable for applications such as seismic monitoring [3] offering a powerful

and versatile solution for detecting acoustic changes over extended distances.

Many works have concentrated on using conventional seismograph generated data for the classification and prediction of earthquake events using deep learning models [4]–[6], however some works have utilized DAS dataset [7] for seismic studies particularly due to its high spatial resolution, long sensing range, and continuous monitoring capability.

In this work, a publicly available dataset comprising distributed acoustic measurements collected from a 41.5 km-long telecom cable deployed offshore Toulon, France was utilized [8]. The dataset includes recordings of a 1.9 magnitude earthquake that occurred approximately 100 km to the northeast of the cable's location. The Least Absolute Shrinkage and Selection Operator (LASSO) algorithm was used to classify earthquake and non-earthquake events, and a comparative analysis with other classifiers was performed to assess the algorithm's efficiency.

The subsequent sections of the paper are structured as follows: Section II provides the methodology, Section III discusses the results obtained and evaluation of the model's performance, and Section IV summarizes the major conclusions drawn from the work.

II. METHODOLOGY

The flowchart representing the overall workflow is illustrated in Fig. 1.

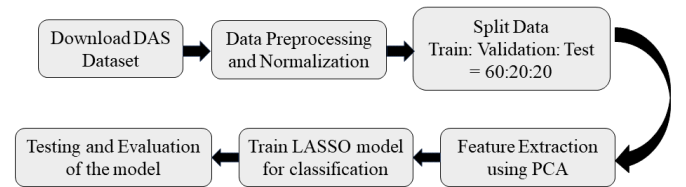


Fig. 1. Methodology flow chart

As illustrated, initially the dataset is downloaded from [9], where it is provided in a .mat file format. The dataset consists of an array of size 6000x6000, where the columns represents the time and rows represent the events. The missing values,

if any is eliminated by filling it with the row's mean value. The data is normalized to the range [0, 1] using Min-Max scaling. After splitting the data, features from the train data are extracted using Principal Component Analysis (PCA). PCA analyzes the variance within each attribute or feature of the dataset and identifies the principal components. These principal components represent the directions in which the data exhibits the highest variability. They are linear combinations of the original attributes and are organized based on the amount of variance they account for. The first principal component captures the most variance, followed by the second principal component, and so forth [10].

Further, the extracted features are used to train the LASSO regression model. It is an extension of the conventional linear regression model, which describes a linear association between features/independent variable and target/dependent variable, given as (3).

$$y = m_0 + m_1x_1 + m_2x_2 + \dots + m_nx_n + \varepsilon \quad (3)$$

where $m_0, m_1, m_2, \dots, m_n$ are the coefficients subject to estimation, $x_0, x_1, x_2, \dots, x_n$ are the features, ε is the error, and y is the target. In LASSO, based on the absolute value of coefficients, the model introduces an additional penalty parameter λ , and hence the L1 regularization term [11] is given by (4).

$$L1 = \lambda \cdot (|m_1| + |m_2| + \dots + |m_n|) \quad (4)$$

The primary objective of LASSO regression is to determine coefficient values that minimize the sum of squared differences between predicted and actual values. Simultaneously, it seeks to minimize the L1 regularization term, which encourages sparsity in the model's coefficients. Thus, the objective function of LASSO model is given as (5).

$$ObjectiveFunction = \sum_{i=1}^k (y_i - \hat{y}_i)^2 + \lambda \cdot \sum_{i=1}^n |m_i| \quad (5)$$

where, $\sum_{i=1}^k (y_i - \hat{y}_i)^2$ indicates the residual sum of squares, that is the sum of the squared differences between the actual target values (y_i) and the predicted values ($\hat{y}_i$), for all data points. It quantifies the error/residuals in the predictions and is commonly used as a measure of the model's goodness of fit in linear regression.

After training the LASSO model with the extracted features, the next step involves feature extraction from both the validation and test sets. The performance of the model is evaluated using a range of metrics, including accuracy, precision, recall, F1 score, and the precision-recall (PR) curve. These metrics collectively provide a comprehensive assessment of the classification capabilities of the model. Furthermore, to assess the efficiency of the LASSO model in earthquake event classification, a comparative analysis is conducted against alternative classifiers, including Support Vector Machine (SVM), Random Forest (RF), K-Nearest Neighbor (KNN), and Decision Tree (DT). This comparative approach facilitates an informed evaluation of the suitability of the model and its

relative performance when compared to established classification methods, thus ensuring the establishment of a robust and reliable earthquake event classification framework.

III. RESULTS AND DISCUSSION

This section presents the outcomes of the earthquake event classification using the DAS dataset and the LASSO model. Subsequently, the comparative evaluation of the LASSO model against alternative classifiers is performed. The analysis reveals remarkable achievements in earthquake event classification.

The LASSO model was evaluated using the performance parameters mentioned in Section II. The values obtained for the metrics are summarized in Table I.

TABLE I
PERFORMANCE METRICS FOR LASSO MODEL

Performance Metrics	Validation Results	Test Results
Accuracy	99.58%	99.58%
Precision	99.05%	99.69%
Recall	99.36%	98.78%
F1 score	99.21%	99.23%

The evaluation of the model also involves the comparison of the model's performance with that of other classifiers, SVM, RF, KNN, and DT. Through this rigorous comparison, the effectiveness and efficiency of the LASSO model are scrutinized in depth. The results of this analysis provide valuable insights into the superior performance of the LASSO model in accurately classifying earthquake events, highlighting its potential as the preferred choice for seismic monitoring and event detection. The bar plots indicating the performance metrics comparison of the five models for validation and testing sets are illustrated in Fig. 2 and Fig. 3, respectively.

The precision-recall curves are meticulously computed for each model, illustrating their ability to distinguish between positive and negative samples. Additionally, area under curve (AUC) values are derived, offering a quantitative measure of model performance. The resulting plot effectively conveys the precision-recall curves for these models, and is shown in Fig. 4.

Notably, the LASSO model exhibited a perfect AUC score of 1.00, demonstrating its remarkable precision and recall capabilities. Additionally, other models, including the DT, KNN, and RF, achieved AUC values hovering around 0.84. These results underscore their robustness and accuracy in distinguishing between the two classes. In contrast, the SVM model displayed a moderate performance with an AUC of 0.70. These findings contribute valuable insights into the comparative performance of these machine learning models, aiding in informed model selection and decision-making in real-world applications.

IV. CONCLUSION

This manuscript sheds light on the potential of DAS for earthquake event classification. Using the dataset collected

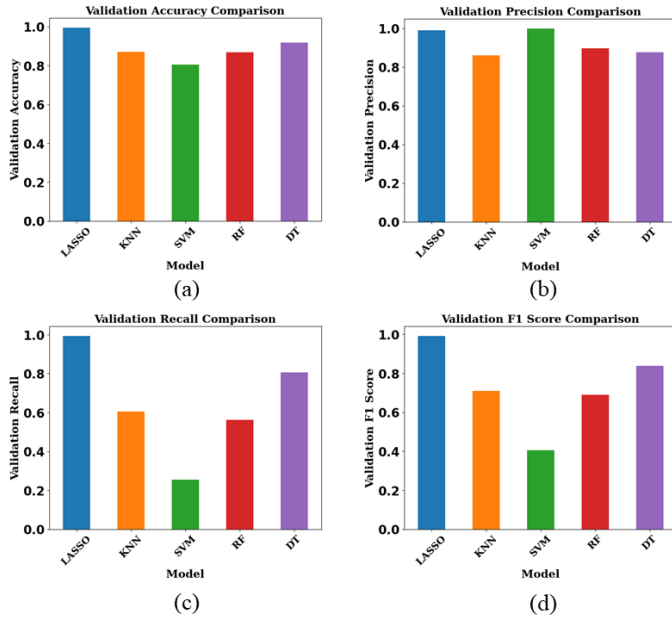


Fig. 2. Performance Comparison of models for Validation Set

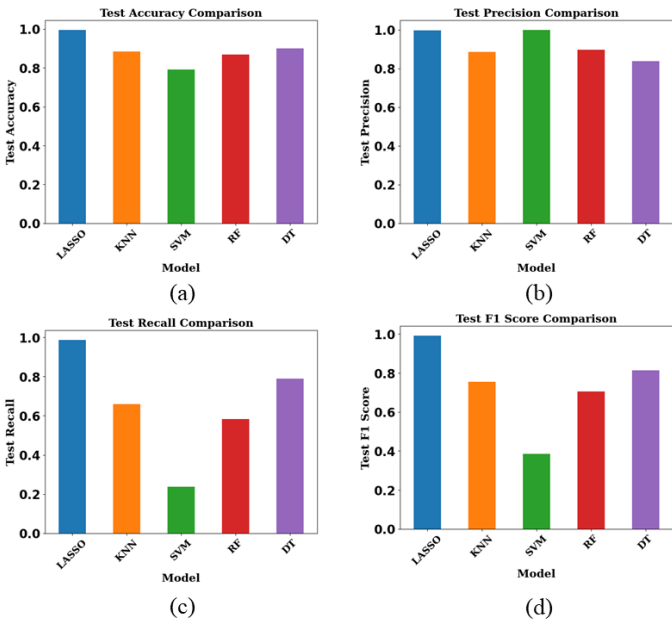


Fig. 3. Performance Comparison of models for Test Set

through DAS technology, the application of the LASSO model resulted in an accuracy rate of 99.58%. Notably, in addition to accuracy, other critical parameters such as precision (99.69%), recall (98.78%), and the F1 score (99.23%) showcase the model's performance in identifying earthquake events.

This performance stands as a testament to the advantages of utilizing DAS data, affirming its position as a robust and effective tool for earthquake event classification. Furthermore, when compared to other contemporary models, the model's accuracy and precision are notably superior. These findings

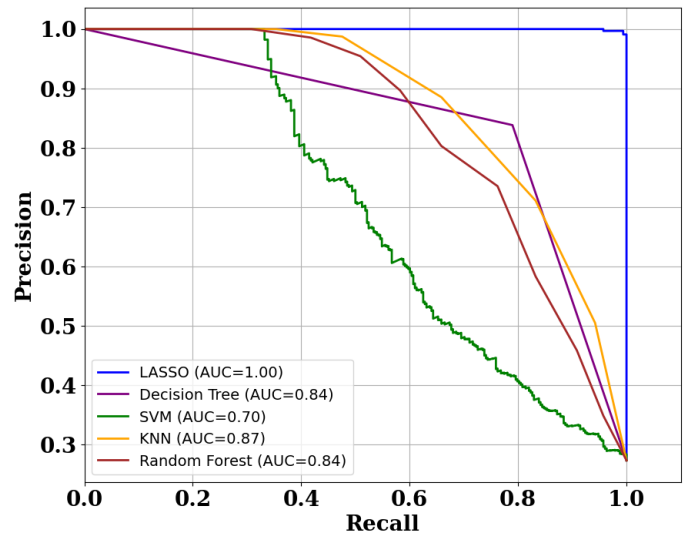


Fig. 4. PR curve and AUC values for the models

emphasize not only the promise of DAS technology in seismic monitoring but also its potential to outperform alternative methods in earthquake detection and characterization. The results underscore the pivotal role that DAS technology can play in advancing the comprehension and response to seismic events in the future.

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